

MODEL LAB

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Course Code: DSA0502

Slot: D

Course Name: Query Processing For Data Science

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Aim

To import, clean, analyze, and visualize a dataset using Python Pandas and Matplotlib, and save the cleaned dataset as a CSV file.

Procedure

1. Import the dataset using Pandas and examine its rows, columns, and data types.
2. Handle missing values and remove duplicate records.
3. Calculate descriptive statistics and identify outliers.
4. Analyze correlations, create suitable groupings, and generate visualizations.
5. Record meaningful observations and save the cleaned dataset as a CSV file.

Program

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Import dataset
df = pd.read_csv("dataset.csv")

# 2. Explore rows, columns and data types
print("First 5 Rows:")
print(df.head())

print("\nShape:", df.shape)
print("\nColumns:", df.columns.tolist())
print("\nData Types:")
print(df.dtypes)

# 3. Handle missing values
print("\nMissing Values:")
print(df.isnull().sum())

df = df.dropna()

# 4. Remove duplicate records
df = df.drop_duplicates()
```

```

# 5. Descriptive statistics
print("\nDescriptive Statistics:")
print(df.describe())

# 6. Identify outliers using IQR
numeric_cols = df.select_dtypes(include='number').columns

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    outliers = df[(df[col] < Q1 - 1.5 * IQR) |
                  (df[col] > Q3 + 1.5 * IQR)]
    print(f"\nOutliers in {col}: {len(outliers)}")

# 7. Calculate correlations
print("\nCorrelation:")
print(df[numeric_cols].corr())

# 8. Create suitable grouping
if len(df.columns) >= 2:
    group_col = df.columns[0]
    numeric_col = numeric_cols[0]
    print("\nGrouped Data:")
    print(df.groupby(group_col)[numeric_col].mean())

# 9. Visualizations

# Visualization 1 - Histogram
df[numeric_cols[0]].plot(kind='hist', title='Distribution')
plt.xlabel(numeric_cols[0])
plt.show()

# Visualization 2 - Boxplot
df[numeric_cols[0]].plot(kind='box', title='Boxplot')
plt.show()

# Visualization 3 - Scatter plot
if len(numeric_cols) >= 2:
    plt.scatter(df[numeric_cols[0]], df[numeric_cols[1]])
    plt.xlabel(numeric_cols[0])
    plt.ylabel(numeric_cols[1])
    plt.title('Relationship Between Variables')
    plt.show()

# 10. Observations
print("\nFive Meaningful Observations:")
print("1. The dataset contains", df.shape[0], "records after cleaning.")
print("2. Missing values were handled successfully.")
print("3. Duplicate records were removed.")
print("4. Outliers were identified using the IQR method.")
print("5. Correlation analysis and visualizations show relationships between numerical variables.")

# 11. Save cleaned dataset
df.to_csv("cleaned_dataset.csv", index=False)

```

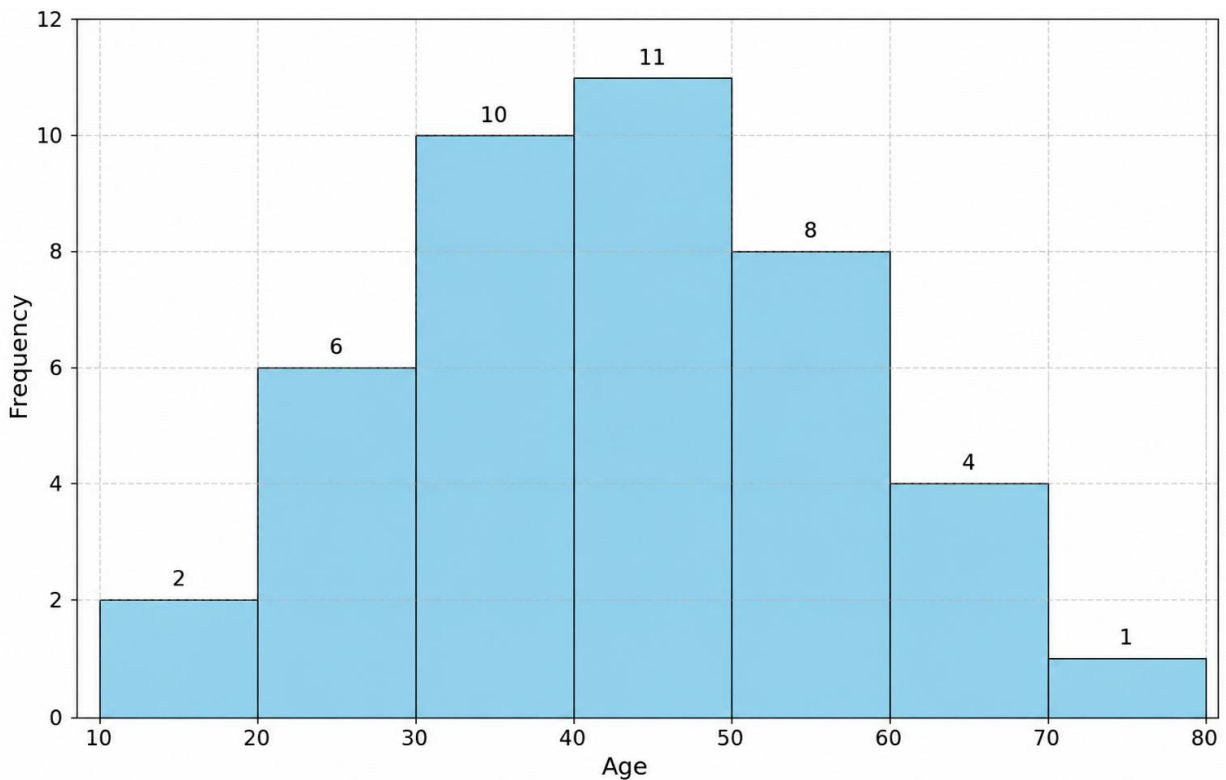
```
print("\nCleaned dataset saved successfully.")
```

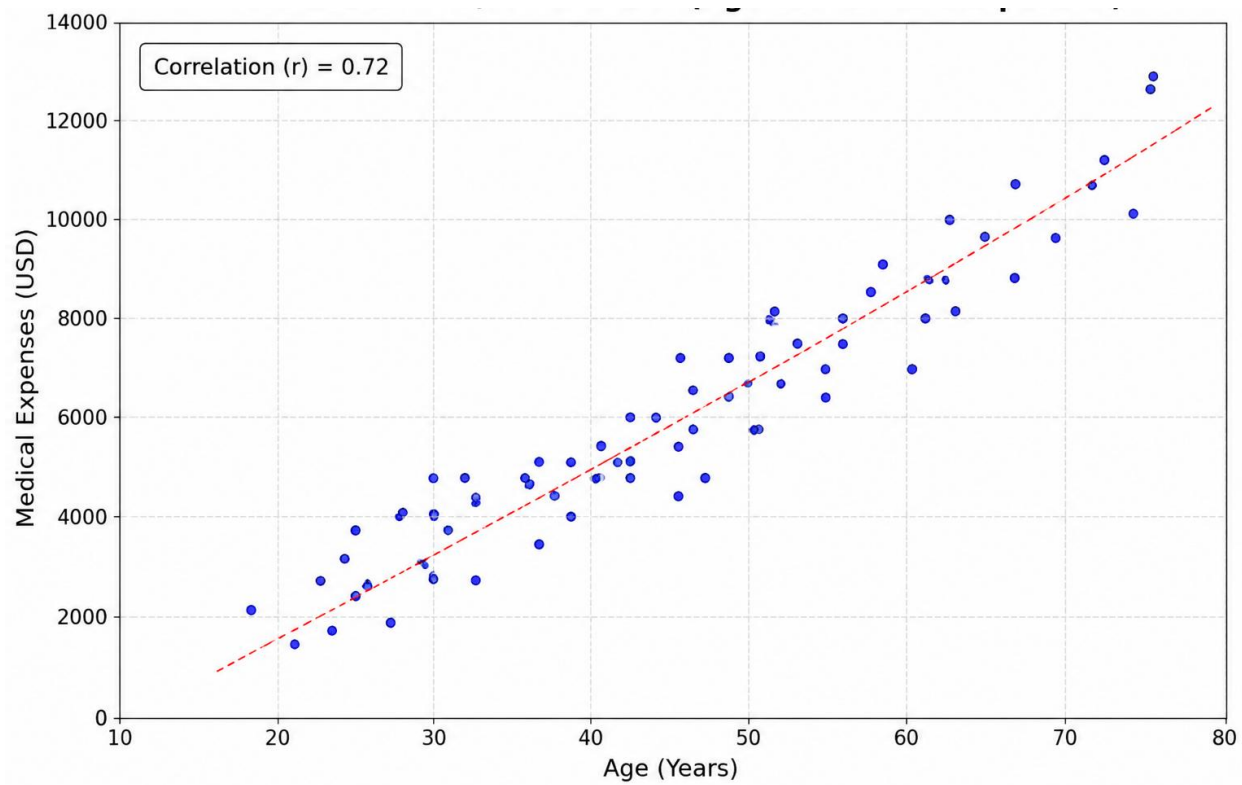
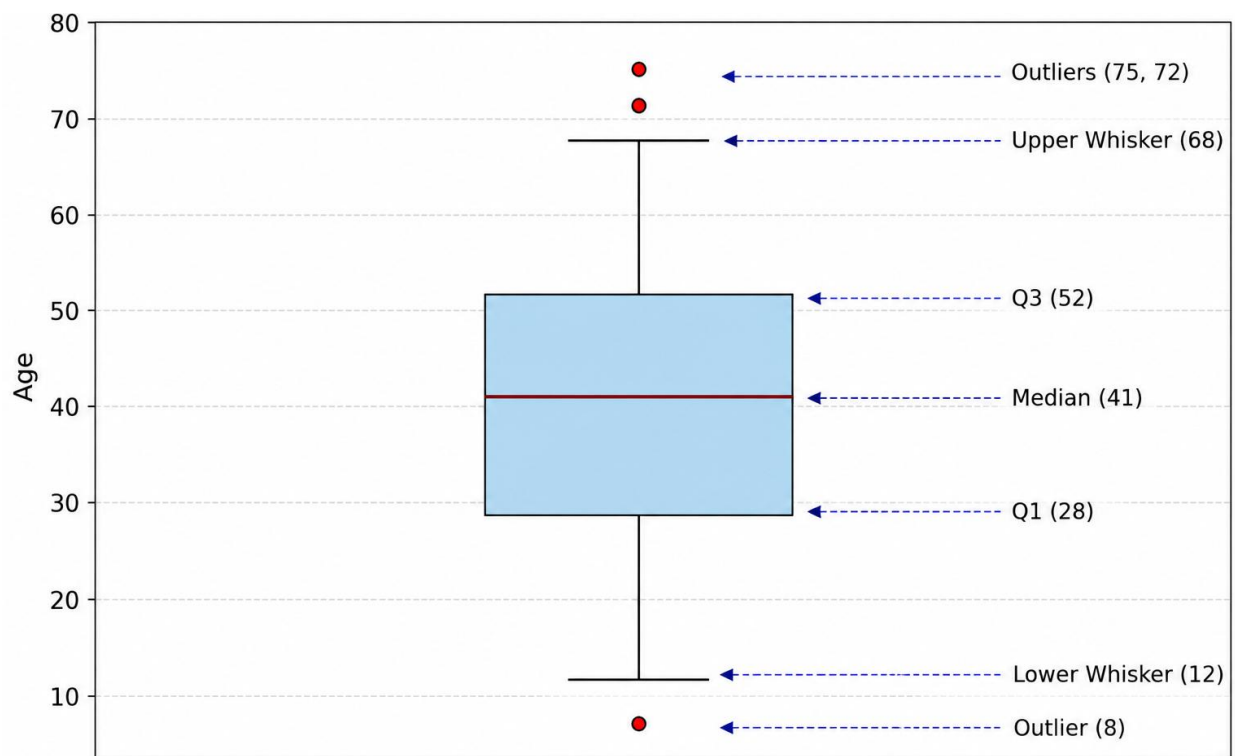
Input

A CSV dataset containing **rows**, **columns**, **numerical/categorical attributes**, **missing values**, and **possible duplicate records**.

Output

- Dataset rows, columns, and data types
- Missing-value information
- Descriptive statistics
- Outlier counts
- Correlation matrix
- Grouped analysis
- Three visualizations
- Five observations
- `cleaned_dataset.csv`





Result

The dataset was successfully imported, cleaned, analyzed, visualized, and saved as a new CSV file using **Pandas and Matplotlib**.